

dft-road-casualty-statistics-casualty-last-5-years Report

Dataset: dft-road-casualty-statistics-casualty-last-5-years.csv

1. Dataset Overview

Item	Value
Number of Rows	5,000
Number of Columns	23
Missing Values	0
Duplicate Rows	0
Target Column	age_of_casualty
Problem Type	Regression

2. Data Quality

Quality Measure	Result
Overall Quality Score	85 / 100
Missing Values	Excellent
Duplicate Rows	Excellent
Data Types	Excellent
Correlation	Poor
Outliers	Fair

3. Target Balance

The selected target classes are reasonably balanced.

Class	Proportion
23	0.0254
19	0.0252
-1	0.0240
18	0.0236
22	0.0234
20	0.0228
28	0.0224
31	0.0224
25	0.0222
30	0.0222
29	0.0220
27	0.0216

24	0.0208
21	0.0206
34	0.0206
36	0.0204
26	0.0202
35	0.0192
33	0.0188
32	0.0188
17	0.0184
38	0.0176
43	0.0160
44	0.0156
49	0.0152
42	0.0152
39	0.0148
45	0.0148
40	0.0148
37	0.0144
52	0.0142
41	0.0142
58	0.0134
56	0.0130
55	0.0128
57	0.0122
54	0.0118
48	0.0114
46	0.0112
50	0.0110
47	0.0110
11	0.0108
59	0.0108
51	0.0104
53	0.0102
15	0.0102
12	0.0102
16	0.0094
60	0.0092
14	0.0092
63	0.0090
61	0.0088
65	0.0082

62	0.0080
13	0.0076
64	0.0072
75	0.0066
76	0.0062
9	0.0062
10	0.0062
66	0.0054
73	0.0054
70	0.0052
78	0.0050
79	0.0050
5	0.0048
4	0.0048
6	0.0044
72	0.0044
68	0.0044
7	0.0042
69	0.0040
67	0.0040
74	0.0040
77	0.0038
71	0.0038
8	0.0038
84	0.0036
80	0.0032
81	0.0026
85	0.0026
83	0.0024
82	0.0018
2	0.0018
86	0.0018
88	0.0016
0	0.0016
87	0.0014
3	0.0014
1	0.0012
90	0.0008
89	0.0006
91	0.0006
93	0.0002

92	0.0002
94	0.0002

4. Machine Learning Results

Model	MAE	RMSE	R2
Random Forest	2.1188	2.6820	0.9813
Linear Regression	3.7549	5.0854	0.9327
Decision Tree	2.8091	3.6973	0.9644

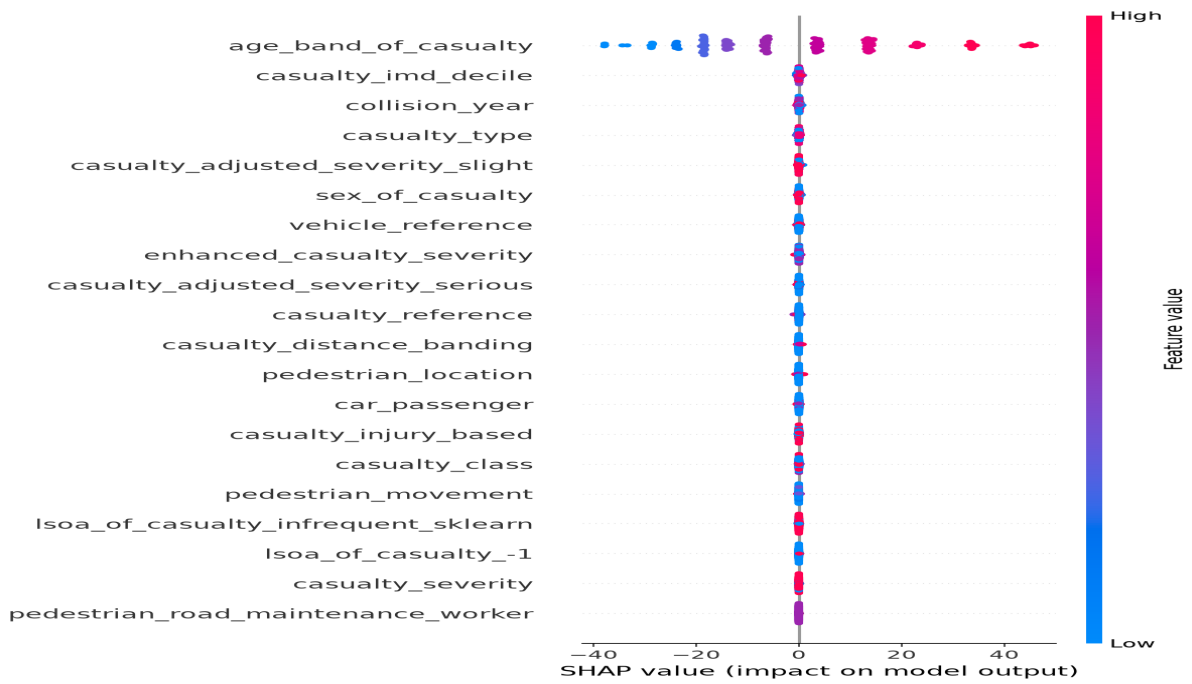
Best Model: Random Forest
R2: 0.9813

5. Explainable AI / SHAP

SHAP was used to explain which features had the strongest influence on the predictions made by the selected best-performing model.

Rank	Feature	Mean SHAP Value
1	age_band_of_casualty	16.4449
2	casualty_imd_decile	0.1591
3	collision_year	0.1160
4	casualty_type	0.1037
5	casualty_adjusted_severity_slight	0.0922
6	sex_of_casualty	0.0847
7	vehicle_reference	0.0740
8	enhanced_casualty_severity	0.0722
9	casualty_adjusted_severity_serious	0.0713
10	casualty_reference	0.0663

SHAP Summary Visualization



6. Conclusion

The selected target column was 'age_of_casualty'. The task was identified as a regression problem. The best-performing model was Random Forest, with a R2 score of 0.9813. The most influential features identified by SHAP were age_band_of_casualty, casualty_imd_decile, collision_year.